

T.VAISHNAVI(192421370)

DEBUGGING -CO2

CSA0312-DATASTRUCTURE

Q1

6. Supply Code in underflow operation.

5 marks

Q2

9. Include missing statements and display the list of elements in linked list.

5 marks

Q3

Identify an error in the following snippet.

5 marks

3/3 answered

SIMATS
ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

THUGUNDRAM VAISHNAVI
19242137008

Q2. Supply Code in underflow operation

```
int pop() {
    if(top == 0) {
        printf("Underflow");
        return -1;
    }
    return stack[top--];
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 int top;
3 int stack[100];
4
5 int pop() {
6     if (top == -1) {
7         printf("Underflow\n");
8         return -1;
9     }
10    return stack[top--];
11 }
12 int main() {
13     top = -1;
14
15     int x = pop();
16     if (x != -1) {
17         printf("Popped element: %d\n", x);
18     }
19     return 0;
20 }
```

Test Input

top = -1

Output

Underflow

Pending manual review

Submitted

Q1

6. Supply Code in underflow operation.

5 marks

Q2

9. Include missing statements and display the list of elements in linked list.

5 marks

Q3

Identify an error in the following snippet.

5 marks

3/3 answered

SIMATS
ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

THUGUNDRAM VAISHNAVI
19242137008

void display(struct Node *head)

```
{
    while(head != NULL){
        printf("%d ", head->data);
        head = head->next;
    }
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 struct Node {
4     int data;
5     struct Node *next;
6 };
7
8 void display(struct Node *head) {
9     while (head != NULL) {
10        printf("%d ", head->data);
11        head = head->next;
12    }
13 }
14
15 int main() {
16     struct Node *head = NULL;
17
18     struct Node *node1 = (struct Node *)malloc(sizeof(struct Node));
19     struct Node *node2 = (struct Node *)malloc(sizeof(struct Node));
20     struct Node *node3 = (struct Node *)malloc(sizeof(struct Node));
21
22     node1->data = 10; node1->next = node2;
23     node2->data = 20; node2->next = node3;
24     node3->data = 30; node3->next = NULL;
25     head = node1;
26     display(head);
27     return 0;
28 }
```

Test Input

stdin input (optional)

Output

10 20 30

Pending manual review

Submitted

QUESTIONS

Q1
6. Buggy Code in underflow operation
5 marks

Q2
9. Include missing statements and display the list of elements in linkedlist
5 marks

Q3
Identify an errors in the following snippet
5 marks

3/3 answered

Q3 Identify an errors in the following snippet

5 marks

```
int binarySearch(int arr[], int n, int key)
{
    int low = 0, high = n - 1;
    while(low < high)
    { int mid = (low + high) / 2;
      if(arr[mid] == key) return mid;
      else if(arr[mid] < key)
        low = mid;
      else high = mid;
    }
    return -1;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
2 int binarySearch(int arr[], int n, int key)
3 {
4     int low = 0, high = n - 1;
5     while (low <= high)
6     {
7         int mid = low + (high - low) / 2;
8
9         if (arr[mid] == key)
10            return mid;
11        else if (arr[mid] < key)
12            low = mid + 1;
13        else
14            high = mid - 1;
15    }
16    return -1;
17 }
18 int main()
19 {
20     int arr[] = {1, 3, 5, 7, 9};
21     int n = 5;
22     int key = 7;
23     int ans = binarySearch(arr, n, key);
24     printf("%d\n", ans);
25     return 0;
26 }
```

Submitted

Test Input

```
arr = [1,3,5,7,9]
n = 5
key = 7
```

Output

```
3
```

Pending manual review